Handwritten Digit Recognition System Using Artificial Neural Networks (ANN)

Project Overview
This project develops an ANN-based system to recognize handwritten digits (0–9) using the MNIST dataset. The application allows users to upload an image, preprocesses it, and predicts the digit with a confidence score through a Flask web application.

Problem Statement

Manual recognition of handwritten digits is slow and error-prone. The objective is to automate digit recognition using an Artificial Neural Network trained on MNIST.

Objectives

• Recognize digits 0–9.
• Train an ANN on MNIST.
• Build a Flask web app.
• Display prediction confidence.
• Achieve >97% accuracy.

Dataset

MNIST contains 70,000 grayscale images (60,000 training and 10,000 testing), each of size 28×28 pixels across 10 classes.

Technology Stack

Frontend: HTML, CSS, Bootstrap, JavaScript
Backend: Python, Flask
ML: TensorFlow, Keras, NumPy, OpenCV, Pillow, Scikit-learn

ANN Architecture

Input (784) → Dense(256, ReLU) → Dropout(0.3) → Dense(128, ReLU) → Dense(64, ReLU) → Output(10, Softmax)

Workflow

Upload Image → Resize (28×28) → Grayscale → Normalize → Flatten → ANN Prediction → Softmax → Display Result

Model Details

Optimizer: Adam
Loss: Categorical Crossentropy
Metrics: Accuracy, Precision, Recall, F1-score, Confusion Matrix

Expected Results

Training Accuracy: ~99%
Testing Accuracy: 97–98%

Applications

Bank cheque recognition, postal ZIP code reading, OMR systems, document digitization, healthcare records.

Future Enhancements

Alphabet recognition, word recognition, real-time webcam support, mobile app deployment, multilingual handwriting recognition, cloud deployment.

Frontend Design (Light Cream Theme)

Background: #FFF8F0, Card: #FFFDF9, Header: #8B5E3C, Buttons: #D4A373, Hover: #BC8A5F, Text: #4E342E. Features include drag-and-drop upload, image preview, rounded cards, soft shadows, confidence indicator, and responsive design.